MILITARY SPECIFICATION

AMPLIFIER, TRAVELING WAVE TUBE

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the performance, design, test, manufacture and acceptance requirements for the traveling wave tube (TWT) amplifier, herein referred to as the amplifier.

1.2 Classification. The amplifier covered by this specification consists of three types:

a. Type I 1.0 to 2.0 Gigahertz (GHz)

b. Type II 2.0 to 4.0 GHz

c. Type III 11.0 to 18.0 GHz

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions, and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, New Jersey 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5865

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:16Z
Check the source to verify that this is the current version before use.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	Preservations, Methods of.
MIL-E-5400	Electronic Equipment, Aerospace, General Specification for.
MIL-T-5422	Testing, Environment, Airborne Electronic and Associated Equipment.
MIL-I-6181	Interference Control Requirements, Aircraft Equipment.
MIL-M-7793	Meter, Time Totalizing.
MIL-E-17555	Electronic and Electrical Equipment Accessories and Repair Parts; Packaging and Packing of.
MIL-T-18303	Test Procedures; Preproduction, Acceptance, and Life for Aircraft Electronic Equipment, Format for.
MIL-N-18307	Nomenclature and Identification for Electronic, Aeronautical, and Aeronautical Support Equipment, Including Ground Support Equipment.
MIL-H-46855	Human Engineering Requirements for Military Systems, Equipment and Facilities.
MIL-C-81703	Connector, Electric, Circular, Miniature, Rack and Panel or Push-Pull Coupling, Environmental Resisting.

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:16Z
Check the source to verify that this is the current version before use.

MIL-STD-130	Identification Marking of U.S. Military Property.
MIL-STD-280	Definitions of Item Levels, Item Interchangeability, Models and Related Terms.
MIL-STD-454	Standard General Requirements for Electronic Equipment.
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment.
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-704	Aircraft Electrical Power Characteristics.
MIL-STD-781	Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution.
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of.
MIL-STD-831	Test Reports, Preparation of.
MIL-STD-882	System Safety Program Requirements.
MIL-STD-965	Parts Control Program.
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-45662	Calibration System Requirements.
MS3102	Connector, Receptacle, Electric, Box Mounting, Solder Contacts, AN Type.
MS3112	Connectors, Receptacle, Electric, Series 1, Box Mounting, Flange, Bayonet Coupling, Solder Contact.
MS17321	Meter, Time Totalizing, Miniature, Digital 28 Volts DC.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:16Z
Check the source to verify that this is the current version before use.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND
(Code Ident 30003)

1356AS235	Amplifier, Radio Frequency AM-7230/DLQ-3C(V)
1356AS236	Amplifier, Radio Frequency AM-7231/DLQ-3C(V)
1356AS237	Amplifier, Radio Frequency AM-7232/DLQ-3C(V)

(Copies of specifications, standards, handbooks, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Item description. The amplifier covered by this specification is a TWT used to amplify radio frequency (RF) signals existing in the microwave region of electromagnetic energy.

3.1.1 Components. The amplifier unit shall, within the meaning of MIL-STD-280, contain the following separate assemblies.

a. TWT assembly.

b. Power supply assembly.

3.2 First article. When specified, a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.3 Interface.

3.3.1 External electrical and RF. The external electrical interfaces for the amplifier shall be provided through connectors J2001, J2005, and J2006, and located as shown on Drawings 1356AS235 for Type I, 1356AS236 for Type II and 1356AS237 for Type III. Connector type and function shall be as specified in Table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:16Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical and RF interface types and functions.

Reference Designation	Connector Type	Pin Assignment	Function
J2001 Input Power/ Control	MS3102E20-8P	A	+28 Vdc
		B	Operate Command (high voltage control)
		C	Remote Standby
		D	28 Vdc Return
		E	Remote Operate
		F	Remote Fault
J2002 Auxiliary (Fan)	Winchester M4SLRGN	A	+28 Vdc
		B	28 Vdc Return
		C	Spare
		D	Spare
J2003 TWT Low Voltage	MS3112E10-6S	A	Ground
		B	Thermostat (NC)
		C	Spare
		D	Anode Voltage
J2004	1/ 2/		High Voltage Lead
J2005	Types I and II - TNC Female Type III-SMA Female		RF Input
J2006	Types I and II - TNC Female, right angle Type III - Waveguide (WR-62)		RF Output
J2007	1/ 3/		High Voltage Lead

1/ Connector type shall be Reynolds 167-8800 (Double Pole) or 167-9183 (Single Pole).
2/ Not used in Type III.
3/ Not used in Types I and II.

Source: https://assist.dla.mil -- Downloaded: 202610-23T11:16Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical and RF interface types and functions. (Continued)

Reference Designation	Connector Type	Pin Assignment	Function
J2008	1/		High Voltage Lead
J2009	1/		High Voltage Lead
J2010	Johnson 105601		Test Jack
J2011	Johnson 105602		Test Jack

3.3.2 External mechanical. The external mechanical interface dimensions shall be as shown on Drawings 1356AS235 for Type I, 1356AS236 for Type II and 1356AS237 for Type III. The mechanical interface shall enable direct installation into Government selected electronic platforms using existing mounting holes.

3.4 Materials, processes, and parts. In the selection of materials, processes, and parts, fulfillment of major design objectives for the intended application specified herein shall be the prime consideration. In doing so, the following shall govern:

a. Microwave technology shall be considered and microwave items shall conform to requirements specified herein.

b. Materials, processes, and parts requirements shall be as specified in MIL-E-5400.

c. Contractor-designed (peculiar) microwave items shall meet the general intent of MIL-E-5400.

d. Approval for the use of nonstandard materials, processes, and parts shall be obtained as specified in MIL-E-5400.

3.5 Characteristics.

3.5.1 Performance requirements. The amplifier shall, when employed in conjunction with other units of the AN/DLQ-3C Electronic Countermeasures Set, meet the performance requirements specified herein at the conditions specified in 4.3. Electron tubes shall be interchangeable between amplifiers of the same type (I, II, or III) with no supply voltage changes.

Source: https://assist.dla.mil -- Downloaded: 202610-23T11:16Z
Check the source to verify that this is the current version before use.

3.5.1.1 Frequency range. The frequency range of the amplifier shall be:

Type I	1.0 to 2.0 Gigahertz (GHz)
Type II	2.0 to 4.0 GHz
Type III	11.0 to 18.0 GHz

3.5.1.2 RF output power. The amplifier, when operated into a 50 ohm load with a voltage standing wave ratio (VSWR) of 1.5:1 ±10 percent varied through all phase angles, shall provide an RF output power as follows:

Type I	100 watts between 1.0 and 2.0 GHz
Type II	100 watts between 2.0 and 4.0 GHz
Type III	80 watts between 11.0 and 12.0 GHz, 100 watts between 12.0 and 17.5 GHz and 80 watts between 17.5 and 18.0 GHz

3.5.1.3 Termination mismatch. The amplifier shall meet specified frequency range and RF output power with an input and output voltage standing wave ratio (VSWR) of 2.0:1 nominal. The amplifier shall not be damaged when operating into load with VSWRs up to and including 2.5:1 over the frequency range and at any phase angle.

3.5.1.4 Input power.

3.5.1.4.1 Input voltage. The amplifier shall operate as required when supplied a power source as specified in Table II and Figures 9 and 10 of MIL-STD-704 except the Steady State Voltage shall be 28.0 ±3.0 Volts direct current (Vdc).

3.5.1.4.2 Current drain. The amplifier current drain shall be not greater than 40.0 amperes (A).

3.5.1.4.3 Low voltage protection. The amplifier shall not be damaged by primary power voltages less than 25.0 Vdc, although specified performance is not required during low-voltage periods. The amplifier shall automatically resume specified operation when the voltage returns within the specified limits.

3.5.1.5 Warm-up time. Within 180 seconds after application of primary power, the amplifier shall achieve specified output power and all other performance requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:16Z
Check the source to verify that this is the current version before use.

3.5.1.6 Small signal gain. The small signal gain shall be greater than 45 dB for the applicable frequency range for each amplifier type.

3.5.1.7 Saturated gain. The saturated gain shall be greater than 40 dB for the applicable frequency range for each amplifier type.

3.5.1.8 Duty cycle. The amplifier shall be capable of 100 percent duty cycle.

3.5.1.9 Spectral purity. Unless otherwise specified in the contract or purchase order (see 6.2.1), all residual noise components within ±100 KHz of the carrier shall be down not less than 25 dB from the carrier. As a design goal, all residual noise components shall be down not less than 30 dB from the carrier.

3.5.1.10 Broadband tube noise. The broadband tube noise shall be not greater than +30 dBm.

3.5.1.11 Thermal protection. Unless otherwise specified in the contract or purchase order (see 6.2.1), the amplifier shall be equipped with automatic, resettable thermal overload protection.

3.5.1.12 Harmonic and spurious emissions. In-band and out-of-band harmonic and spurious emissions shall be suppressed not less than 30 dB from the carrier level.

3.5.1.13 Fault protection. The amplifier shall contain electronic overload and thermal cut-out and time-delay circuits to provide fault protection for both microwave tube and power supply. The protective functions shall automatically reset upon clearance of the fault. The amplifier shall provide accessible test points for microwave tube helical current and voltage functions.

3.5.1.14 Controls and indicators. The amplifier shall provide the following controls and status indicators:

a. Operate command. The operate command shall be +25.0 to 31.0 Vdc supplied by the external system and shall cause application of high voltage to the traveling wave tube (TWT). The operate command may be applied coincident with application of input primary power.

b. Status monitoring. Remote monitoring of Standby, Operate, and Fault conditions shall be indicated by presence of +5 ±0.5 Vdc, 20 milliampere maximum at connector J2001 as specified in Table I.

Source: https://assist.dla.mil -- Downloaded: 202640-23T11:16Z
Check the source to verify that this is the current version before use.

3.5.2 Physical characteristics.

3.5.2.1 Conformance to documents. The amplifier shall be as specified in Drawings 1356AS235 for Type I, 1356AS236 for Type II and 1356AS237 for Type III and shall meet all the requirements of this specification and the documents referenced to the extent specified herein.

3.5.2.2 Dimensions. The length, width, and height of the amplifier shall be as specified in Drawings 1356AS235 for Type I, 1356AS236 for Type II, and 1356AS237 for Type III.

3.5.2.3 External adjustments. No external adjustments shall be available on the amplifier.

3.5.2.4 Weight. The weight of the amplifier shall be not greater than 38 pounds.

3.5.2.5 Transportation and storage. Amplifiers shall be capable of being transported by normally available common commercial carriers (air and land) and military carriers (land, sea and air) without damage or adverse impact on operating characteristics. Shipping and storage containers/packaging shall be adequate for storage in facilities normally available to U.S. military units.

3.6 Environmental conditions. The amplifier shall meet all requirements specified herein prior to, during, and after the following environmental conditions, separately or any combination thereof. The amplifier shall not exhibit any temporary or permanent degradation as the result of exposure to any of the following environments:

a. Temperature-altitude: Barometric pressure reduced to the equivalent of 50,000 feet altitude and a temperature of -40°C.

b. Vibration: Random vibration at 10 to 2000 Hz and ±10 g RMS.

c. Shock: Acceleration pulses at 25 g levels with a duration of 11 ms in all axes in any direction.

d. Humidity: An atmosphere containing 85 percent or greater relative humidity at temperature ranging from +28°C to +71°C.

e. Explosion: An explosive atmosphere containing a computed air/fuel ratio of 13:1 at 10,000 feet altitude increments from sea level to 50,000 feet.

f. Storage temperature: Temperatures ranging from -54°C to +71°C.

Source: https://assist.dla.mil -- Downloaded: 202610-23T11:16Z
Check the source to verify that this is the current version before use.

g. Operating temperature: Temperatures ranging from -40°C to +71°C.

h. Thermal shock: Temperatures ranging from -54°C to +71°C.

3.7 Shelf life. The amplifier shall be designed to function and operate as required herein following a storage period of 5 years in normally available Government storage facilities at temperatures varying from -54°C to +71°C and relative humidities varying from 5 percent to 95 percent.

3.8 Reliability. The amplifier shall have a mean-time-between-failure (MTBF) of not less than 100 hours when tested as specified in MIL-STD-781, Test Level F (modified to limit testing between -40°C and +71°C) and using the acceptance criteria specified in Test Plan V-C.

3.9 Design and construction.

3.9.1 Materials. All materials used in the construction of the amplifier shall be as specified in MIL-STD-454, Requirement 3, Flammable Materials; Requirement 4, Fungus-Inert Materials; Requirement 15, Ferrous Alloys, Corrosion Resistance; and Requirement 16, Dissimilar Metals.

3.9.2 Electromagnetic interference. Unless otherwise specified in the contract or purchase order (see 6.2.1), the amplifier shall not generate EMI signals either through radiation, power line conduction, amplitude or phase modulation of the output signal nor through spikes fed back to inputs selected or not selected. Shielding of the amplifier from stray or operating environment EMI shall be provided as specified in MIL-I-6181 and MIL-STD-461.

3.9.3 Nameplate and marking. The nameplate shall be as specified in Drawings 1356AS235 for Type I, 1356AS236 for Type II, and 1356AS237 for Type III. Interchangeable parts and assemblies shall be legibly marked. All markings shall be as specified in MIL-STD-130.

3.9.3.1 Serial numbers. The contractor shall prepare requests for serial number assignments as specified in MIL-N-18307 (see 6.2.2).

3.9.4 Interchangeability. The amplifier unit, with separately attached or integrated power supply assembly, shall be interchangeable as specified in MIL-STD-454, Requirement 7. Electrical and electronic component parts shall conform to the requirements of MIL-STD-965.

3.9.5 Safety. The amplifier produced as specified in this specification shall present no safety hazard to operating personnel. Provisions of MIL-STD-882 shall be complied with.

Source: https://assist.dla.mil -- Downloaded: 202640-23T11:16Z
Check the source to verify that this is the current version before use.

3.9.6 Human performance/human engineering. The amplifier shall be constructed and configured to minimize human error during installation, operation, removal and maintenance. MIL-STD-1472 and MIL-H-46855 shall be used by the contractor to meet human engineering and performance requirements.

3.9.7 Standards of manufacture.

3.9.7.1 Primary DC power connector. The primary DC electrical power connector for the amplifier shall be as specified in MIL-C-81703, Class E and shall mate with MS3102 connector.

3.9.7.2 RF coaxial connector. The amplifier RF coaxial connectors shall be front bulkhead mounted, female receptacle. Type of connectors shall be as identified in Table I.

3.9.7.3 Elapsed time indicator. The amplifier shall incorporate an elapsed time indicator as specified in MIL-M-7793, Type MS17321-10.

3.9.7.4 Power time delay. The amplifier shall incorporate a time delay feature for application of high voltages to the TWT.

3.10 Soldering practices. Soldering practices shall be as specified in MIL-STD-454, Requirement 5.

3.11 Workmanship. The amplifier, including all parts and accessories, shall be constructed and finished in a manner that will ensure compliance with all requirements of this document. Unless otherwise specified herein, fabrication and assembly practices shall be as specified in MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The contractor shall assure product conformance to the requirements, inspections, and tests specified herein. The contractor's quality assurance program shall be planned and used in a manner to achieve levels of quality and reliability commensurate with the intended

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:16Z
Check the source to verify that this is the current version before use.

application specified herein. The contractor shall notify the procuring activity of planned testing not less than 10 days prior to testing.

4.1.1.1 Process control procedures. Process control procedures shall be prepared when necessary to supplement applicable process specifications to provide detailed performance and control methods. These procedures shall document the preparation; fabrication details; conditions to be maintained during each phase of the process; the methods of verifying the adequacy of processing materials, solutions, equipment and their associated control parameters, including statistical quality control plans where applicable and the required records to indicate the results of such inspection and process verification. The contractor's quality organization shall review the written procedures for process controls and conduct audits to determine that the actual operations conform with approved methods and procedures.

4.1.2 Test plans. Unless otherwise specified in the contract or purchase order (see 6.2.1), detailed test plans as specified in MIL-T-18303 shall be prepared by the contractor prior to the commencement of testing (see 6.2.2).

4.1.3 Test data. Unless otherwise specified in the contract or purchase order (see 6.2.1), the detailed test data from both the first article tests and acceptance tests shall be prepared by the contractor (see 6.2.2). Test data for each amplifier offered for acceptance shall be as specified in MIL-STD-831. This data shall identify all rejections from acceptance tests and corrective actions taken.

4.1.4 Failure records. Unless otherwise specified in the contract or purchase order (see 6.2.1), failure analysis and proposed corrective action shall be prepared by the contractor (see 6.2.2).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspections (see 4.5).

b. Quality conformance inspections (see 4.6).

1. Acceptance tests (see 4.6.1).

2. Reliability tests (see 4.6.2).

3. Manufacturing burn-in test (see 4.6.3).

4.3 Inspection conditions. Unless otherwise specified in the applicable test procedure, all tests and inspections shall be conducted under the following conditions:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:16Z
Check the source to verify that this is the current version before use.

a. Temperature (ambient): +23° ±10°C

b. Atmospheric pressure: 725 +50/-115 mm of mercury (28.5 +2.0/-4.5 inches of mercury)

c. Humidity: 10 to 90 percent

4.3.1 External power. External power shall be +28.0 ±3.0 Vdc applied to J2001 and shall simulate airborne vehicle power transfer, when specified in the applicable test procedure.

4.3.2 Test equipment accuracy. Unless otherwise specified herein, all test equipment used shall have an accuracy within 20 percent of the tolerance for the variable to be measured. In some cases where an open-ended minimum or maximum requirement is given, the test equipment used shall be accurate to within 5 percent of that requirement.

4.3.2.1 Maintenance and calibration. All test equipment shall be maintained and calibrated as specified in MIL-STD-45662. Certification to this effect prior to all inspections and tests shall be prepared by the contractor (see 6.2.2).

4.4 First article sample. Unless otherwise specified in the contract or purchase order (see 6.2.1), a first article sample consisting of two amplifiers shall be delivered to a designated testing activity specified in the contract or purchase order (see 6.2.1), for first article inspection. The first article sample shall be manufactured using the same methods, materials, processes and procedures proposed for production.

4.4.1 First article approval. Approval of the first article sample shall be by the procuring activity upon satisfactory completion of all tests. The procuring activity reserves the right to require an additional sample and test if any amplifier fails to meet any requirement specified herein (see 6.2.1). Any production by the contractor before acceptance of the first article sample will be at the contractor's risk. Subsequent to first article approval, the contractor shall not change materials, processes, or procedures without prior approval of the procuring activity (see 6.2.2).

4.5 First article tests. The first article sample shall be subjected to the inspections and tests of Table II and shall be performed in the sequence specified therein. The contractor shall prepare and include test reports (see 6.2.2).

4.5.1 First article failure analysis and corrective action. Unless otherwise specified in the contract or purchase order (see 6.2.1), the procuring

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:16Z
Check the source to verify that this is the current version before use.

TABLE II. First article and quality conformance tests.

Examination or Test	Requirement Paragraph	Method Paragraph	
		First Article	Acceptance
1. Visual examinations			
a. Dimensions	3.5.2.2	4.7.1.1	4.7.1.1
b. Weight	3.5.2.4	4.7.1.1	4.7.1.1
c. Nameplate and markings	3.9.3	4.7.1.3	4.7.1.3
d. Interchangeability	3.9.4	4.7.1.4	4.7.1.4
e. Workmanship	3.10, 3.11	4.7.1.5	4.7.1.5
2. Manufacturing burn-in test		4.6.3	4.6.3
3. Performance			
a. Warm-up and operational stability	3.5.1.5	4.7.2.1	4.7.2.1
b. Termination mismatch	3.5.1.3	4.7.2.2	
c. Low voltage operation	3.5.1.4.3	4.7.2.3	
d. Fault protection	3.5.1.13	4.7.2.4	
e. Controls and indicators	3.5.1.14, 3.9.7.3	4.7.2.5	
f. Primary power voltage variation	3.5.1.4.1	4.7.2.9	4.7.2.9
g. Electromagnetic interference	3.9.2	4.7.2.10	
4. Environmental	3.6	4.7.3	
5. Reliability	3.8	4.6.2	

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:16Z
Check the source to verify that this is the current version before use.

activity shall be notified within one working day of a first article test failure and its probable cause. Procuring activity approval is required before implementation of any corrective action in first article products (see 6.2.2).

4.5.2 Disposition of first article test sample. Unless otherwise specified in the contract or purchase order (see 6.2.1), amplifiers which have been subjected to the first article tests shall be reworked by the contractor by replacing all worn or damaged items. After reworking, the contractor shall resubmit the amplifiers for acceptance.

4.6 Quality conformance tests.

4.6.1 Acceptance tests. Acceptance tests shall be conducted on each amplifier offered for acceptance and shall consist of the tests specified in Table II.

4.6.1.1 Acceptance test data. The contractor shall prepare acceptance test data sheets (see 6.2.2) for each amplifier offered for acceptance.

4.6.2 Reliability tests. Test plan V-C specified in MIL-STD-781 shall be used to determine acceptance or rejection of the reliability test. Theta (θ) shall be 100 hours. All time accumulated in 4.6.1 shall be counted in determining whether an accept or reject condition exists. The contract or purchase order shall specify whether, upon equipment failure prior to reaching an accept or reject decision, the selected first article unit or units shall be repaired to the satisfaction of the Government and subjected to further reliability testing or whether a subsequent first article unit or units shall be randomly selected (see 6.2.1). The contractor shall prepare a reliability test plan, test reports, and a failure summary and analysis report (see 6.2.2).

4.6.2.1 Reliability failure criteria. In addition to the failure criteria specified in MIL-STD-781, an amplifier shall be considered to have failed when any of the monitored characteristics perform outside the specified limits for that characteristic. Preventive maintenance measures or adjustments shall not be performed upon the amplifiers during the period of the test.

4.6.2.2 Reliability data and test reports. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor shall prepare reliability test data and reports, including failure records, for each amplifier subjected to reliability testing (see 6.2.2). The data and reports shall be as specified in MIL-STD-781 and shall be prepared for review not later than 10 working days after completion of reliability testing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:16Z
Check the source to verify that this is the current version before use.

4.6.3 Manufacturing burn-in test. The amplifier shall be operated under the conditions specified herein for a period of ten hours without failure. A failure shall be anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the amplifier. This test shall be deleted if the reliability test includes a test on each amplifier which requires at least ten hours of amplifier on time. The test shall be run at ambient temperature and humidity conditions. Vibration shall be at any non-resonant frequency between 20 and 30 Hz at a level of ±3 gs. The amplifier shall be vibrated for a period of ten minutes prior to the beginning of the ten hour period of operation. Where feasible, the amplifier shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for five minutes and lateral to the plane for five minutes. Where it is not feasible to vibrate the amplifier in two directions, the vertical direction shall be used. During the ten hour period of operation following the ten minute vibration period, the amplifier shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the ten minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future amplifiers. A record shall be kept of all failures. The ten hour period specified above may be composed of two five hour periods to conform with standard working hours.

4.7 Inspection and test methods. Details of methods and procedures shall be as specified in the approved test plan (see 4.1.2).

4.7.1 Visual examinations.

4.7.1.1 Physical characteristics. The amplifier shall be inspected for conformance with 3.5.2.

4.7.1.2 Materials, processes, and parts. The amplifier shall be inspected for conformance with 3.4.

4.7.1.3 Nameplate and marking. The amplifier shall be inspected for conformance with 3.9.3.

4.7.1.4 Interchangeability. The amplifier shall be examined for conformance with 3.9.4.

4.7.1.5 Workmanship. The amplifier shall be examined for conformance with 3.10 and 3.11.

Source: https://assist.dla.mil -- Downloaded: 202640-23T11:16Z
Check the source to verify that this is the current version before use.

4.7.2 Performance tests. Unless otherwise specified in the contract (see 6.2.1), performance tests shall be performed as specified in Table II. The test specimen shall be operated for conformance with 3.5.1. When tests are conducted with no incremented or swept input frequency applied, the input signal shall be at midband sufficient to cause rated saturated output and the product load shall be a mismatched load with a VSWR of 2.0:1 ±10 percent with respect to 50-ohms and worst case phase angle adjusted. Input power shall be as specified in 3.5.1.4. The performance observations specified in 4.7.2.1 through 4.7.2.3 shall be continuously monitored during the conduct of test methods 4.7.2.4 through 4.7.2.8.

4.7.2.1 Warm-up and operational stability. The amplifier shall be operated under the conditions specified in 4.7.2. The input power applied shall be as specified in 3.5.1.4 and the warm-up time shall be as specified in 3.5.1.5. After preconditioning, the amplifier shall be operated for a period of 30 minutes. Beginning at input power turn-on and during the test period the following shall be observed:

a. Frequency range (see 4.7.2.7).
b. RF output power (see 4.7.2.6).
c. Small signal gain (see 4.7.2.8).
d. Saturated gain (see 4.7.2.7).
e. Spectral purity (see 4.7.2.11).
f. Broadband tube noise (see 4.7.2.12).
g. Harmonic and spurious emissions (see 4.7.2.13).

4.7.2.2 Termination mismatch. The amplifier shall be connected to a mismatched load having a VSWR of 2:1 (with phase angle between the resistive and the reactive load components varied between zero and 360 degrees). The signal input shall be at midband and input power shall be adjusted to +25.0 -0.0, +0.5 volts dc and the following observed:

a. Frequency range (see 4.7.2.7).
b. RF output power (see 4.7.2.6).
c. Broadband tube noise (see 4.7.2.12).

The amplifier shall then be connected to a load having a VSWR of 2.5:1 with the phase angle varied between zero and 360 degrees and the following observed:

d. RF output may be less than as in 3.5.1.2.
e. The amplifier load mismatch test shall be repeated except that input power shall be adjusted to +31.0 -0.5, +0.0 volts dc.

4.7.2.3 Low voltage operation. The amplifier shall be operated under the conditions specified in 4.7.2 at midband. The input power shall be reduced

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:16Z
Check the source to verify that this is the current version before use.

to +16 ±0.5 volts dc for a period of not less than five minutes. At the end of this period, the input power shall be restored within the limits specified in 3.5.1.4.1 and the following observed:

a. Frequency range (see 4.7.2.7).
b. RF output power (see 4.7.2.6).
c. Spectral purity (see 4.7.2.11).

4.7.2.4 Fault protection. The amplifier shall be tested for fault protection by application of non-damaging simulated faults. The operation of fault protection functions and the associated presence and non-presence of fault indicators shall meet the requirements specified in 3.5.1.13.

4.7.2.5 Controls and indicators. The amplifier controls and indicators shall be tested under simulated operational conditions with running time observed and meet the requirements specified in 3.5.1.14 and 3.9.7.3.

4.7.2.6 Gain at rated RF output power. The amplifier shall be operated under the conditions of 4.7.2 except that an input signal sufficient to produce rated RF output power shall be swept from:

Type I - 1.0 to 2.0 GHz at increments of 0.25 GHz

Type II - 2.0 to 4.0 GHz at increments of 0.25 GHz

Type III - 11.0 to 18.0 GHz at increments of 1.0 GHz

The amplifier input signal power and the amplifier RF output shall meet the requirements specified in 3.5.1.2.

4.7.2.7 Frequency range and saturated power output. The amplifier shall be operated under the conditions of 4.7.2 except that an input signal sufficient to cause saturated operation shall be swept from:

Type I - 1.0 to 2.0 GHz at increments of 0.25 GHz

Type II - 2.0 to 4.0 GHz at increments of 0.25 GHz

Type III - 11.0 to 18.0 GHz at increments of 1.0 GHz

The amplifier output shall meet the requirements specified in 3.5.1.2.

4.7.2.8 Small signal gain. The amplifier shall be operated under the conditions of 4.7.2 except that an input signal sufficient to produce RF output power 10 dB below rated saturation shall be swept from:

Source: https://assist.dla.mil -- Downloaded: 202640-23T11:16Z
Check the source to verify that this is the current version before use.

Type I - 1.0 to 2.0 GHz at increments of 0.25 GHz

Type II - 2.0 to 4.0 GHz at increments of 0.25 GHz

Type III - 11.0 to 18.0 GHz at increments of 1.0 GHz

The amplifier RF output power shall meet the requirements specified in 3.5.1.6.

4.7.2.9 Primary power voltage variation.

a. Steady-state voltage test: With an input signal of 2.0 GHz for Type I, 4.0 GHz for Type II and 18.0 GHz for Type III sufficient to cause rated saturated RF output power from the amplifier, primary power shall be applied. The primary power shall be adjusted to +25.0, -0.0, +0.5 volts dc and the following observed:

1. Frequency range (see 4.7.2.7).
2. RF output power (see 4.7.2.6).

With a primary power voltage of +31.0, -0.5, +0.0 volts dc applied, the test and observations shall be repeated.

b. Transient voltage test: The amplifier shall be operated with a primary input voltage of +28.0 ±3.0 volts dc for a period of three minutes while monitoring RF output power and frequency range. The primary input voltage shall be reduced to zero for not less than 1 second, raised to +60 ±1.0 volts dc for not more than 100 ms, reduced to -60 ±1.0 volts dc for not more than 100 ms, then returned to +28.0 ±3.0 volts dc and the following observed:

1. Frequency range (see 4.7.2.7).
2. RF output power (see 4.7.2.6).

c. Primary power ripple test: DC primary power has superimposed ripple with an amplitude of 3.0 volts rms and with a frequency spectrum from 50 to 15,000 Hz shall be applied to the amplifier and the following observed:

1. Frequency range (see 4.7.2.7).
2. RF output power (see 4.7.2.6).
3. Spectral purity (see 4.7.2.11).

Source: https://assist.dla.mil -- Downloaded: 202640-23T11:16Z
Check the source to verify that this is the current version before use.

4.7.2.10 Electromagnetic interference. The power lead conducted emission test CE-01 and the radiated emission electric field test RE-01 specified in MIL-STD-462 shall be performed with the amplifier operating under the conditions specified in 4.7.2. The amplifier shall be operated, as specified herein, and continue to meet the requirements specified in 3.9.2. Unless otherwise specified in the contract or purchase order (see 6.2.1), EMI test results shall be prepared by the contractor for each production lot (see 6.2.2).

4.7.2.11 Spectral purity. The amplifier shall be operated at saturation with an RF input signal applied between:

Type I - 1.0 to 2.0 GHz at increments of 0.25 GHz

Type II - 2.0 to 4.0 GHz at increments of 0.25 GHz

Type III - 11.0 to 18.0 GHz at increments of 1.0 GHz

The measurement resolution shall be 10 KHz per division. The spectral purity shall meet the requirements specified in 3.5.1.9.

4.7.2.12 Broadband tube noise. The amplifier shall be operated with no RF input signal applied and shall meet the requirements specified in 3.5.1.10.

4.7.2.13 Harmonic and spurious emissions. The amplifier shall be operated at saturation with an RF input signal applied between:

Type I - 1.0 to 2.0 GHz at increments of 0.25 GHz

Type II - 2.0 to 4.0 GHz at increments of 0.25 GHz

Type III - 11.0 to 18.0 GHz at increments of 1.0 GHz

The amplifier output shall meet the requirements specified in 3.5.1.12 across the frequency range of 1.0 to 18.0 GHz.

4.7.3 Environmental tests. The amplifier shall conform to requirements for Class I equipments operating in environments specified in 3.6 when subjected to testing as specified in Table II of MIL-T-5422, except that salt-fog testing is not required.

4.7.3.1 Operational environmental tests. The amplifier's performance shall be tested and meet the requirements specified in 3.5.1 during and after exposure to the following environments to verify conformance specified in 3.6.

Source: https://assist.dla.mil -- Downloaded: 202640-23T11:16Z
Check the source to verify that this is the current version before use.

4.7.3.1.1 Temperature and altitude test. The amplifier shall be tested in accordance with the temperature-altitude test specified in MIL-T-5422 Table II, as modified herein by Table III.

4.7.3.1.2 Vibration test. The amplifier shall be tested in accordance with the vibration test specified in MIL-T-5422, Curve I and Curve II of Figure 5.

4.7.3.1.3 Shock test. The amplifier shall be tested in accordance with the shock test specified in MIL-T-5422, except as modified herein to no greater than 25 g.

TABLE III. Modified MIL-T-5422 temperature-altitude test.

Step	Temperature (degree C)	Altitude (feet)	Time (hours)	Input Voltage	Operating/ Non-operating
1	-40	ATM 1/	2	--	N.O.
2	-35	ATM	--	25.0	OP
3	-40	50,000	--	31.0	OP
4	-10	ATM	--	31.0	OP
5	+71	ATM	--	--	N.O.
6	+71	ATM	2	31.0	OP
7	OMIT				
8	OMIT				
9	+44	50,000	1.25	31.0	OP
9A	AMB 2/	AMB	2	--	N.O.
9C	0	50,000	1.25	31.0	OP
10	+58	35,000	2	31.0	OP
11	OMIT				
12	OMIT				
13	OMIT				

1/ ATM equals atmosphere
2/ AMB equals ambient

4.7.3.1.4 Explosion test. The amplifier shall be tested in accordance with the explosion test specified in MIL-T-5422.

4.7.3.2 Non-operational environmental tests. The amplifier performance shall be tested and meet the requirements specified in 3.5.1 after exposure to the following environments.

4.7.3.2.1 Thermal shock test. The amplifier shall be tested in accordance with the thermal shock test specified in MIL-T-5422.

Source: https://assist.dla.mil -- Downloaded: 202640-23T11:16Z
Check the source to verify that this is the current version before use.

4.7.3.2.2 Humidity test. The amplifier shall be tested in accordance with the humidity test specified in MIL-T-5422, except that two continuous cycles shall be used instead of 10.

4.8 Pre-shipment inspection. Prior to shipment, inspection shall ensure that preservation, packaging, packing, and marking for shipment are in accordance with the requirements specified in Section 5.

5. PACKAGING

5.1 Preservation-packaging, and packing. Unless otherwise specified in the contract or purchase order (see 6.2.1), the level of preservation-packaging, and packing shall be as specified in MIL-E-17555 and MIL-STD-794, Level C. The method of preservation for Level C shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794 and the method of packaging shall be as specified in MIL-P-116.

5.2 Marking.

5.2.1 Special marking. Special marking shall be as specified in the contract or purchase order (see 6.2.1).

5.2.2 Normal marking. Unit packages and shipping containers shall be marked as specified in the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The amplifier is an operating component of an electronic warfare (EW) countermeasures unit for installation in aerial targets and piloted aircraft.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Spectral purity, if different than specified (see 3.5.1.9).

c. Thermal protection, if different than specified (see 3.5.1.11).

d. Electromagnetic interference, if different than specified (see 3.9.2).

e. Responsibility for inspection (see 4.1).

Source: https://assist.dla.mil -- Downloaded: 202640-23T11:16Z
Check the source to verify that this is the current version before use.

f. Test plans, if different than specified (see 4.1.2).

g. Test data, if different than specified (see 4.1.3).

h. Requirement for failure records, if different than specified (see 4.1.4).

i. First article sample, if different than specified (see 4.4).

j. Designated testing activity for first article inspection (see 4.4).

k. Additional samples and tests, if required (see 4.4.1).

l. Notification time of first article failure, if different than specified (see 4.5.1).

m. Disposition of first article sample, if different than specified (see 4.5.2).

n. Repair or selection of new first article sample for reliability testing (see 4.6.2).

o. Reliability data and test reports, if different than specified (see 4.6.2.2).

p. Performance tests, if different than specified (see 4.7.2).

q. EMI test results, if different than specified (see 4.7.2.10).

r. Level of preservation, packaging, and packing, if different than specified (see 5.1).

s. Special markings, if required (see 5.2.1).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9(n)(2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:16Z
Check the source to verify that this is the current version before use.

Paragraph No.	Data requirement title	Applicable DID No.	Option
3.9.3.1	Requests, Nomenclature, and Identification Assignment and Approval	UDI-E-23090B	
4.1.2	Test Procedures	UDI-T-21347	
4.6.1.1	Records, Inspection and Test	DI-T-22706A	
4.1.3, 4.1.4, 4.5.1	Failure Analysis and Corrective Action Report	DI-R-5299C	
4.3.2.1	Equipment Calibration Procedures	DI-R-7065	
4.4.1	Engineering Change Proposals (ECPs), Requests for Deviations and Waivers (Long Form)	DI-E-2037	
4.1.3, 4.5	Report, First Article Test	UDI-T-23790	
4.6.2	Plan, Reliability Test	DI-R-7033	
4.6.2, 4.6.2.2	Reports, Reliability Test, and Demonstration	DI-R-7034	
4.6.2	Report, Failure Summary and Analysis	DI-R-7041	
4.7.2.10	Electromagnetic Interference Test Report	DI-R-7062	

(Data item descriptions related to this specification, and identified in section 6, will be approved and listed as such in DOD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of two amplifiers. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for examinations, test and approval of the documents first article.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:16Z
Check the source to verify that this is the current version before use.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing Activity

Navy - AS

(Project 5865-N042)

☆U.S. GOVERNMENT PRINTING OFFICE: 1993—705-040/5406

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:16Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY
Commanding Officer
Naval Air Engineering Center
Engineering Specifications and Standards Department
(ESSD), Code 93
Lakehurst, NJ 08733

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 12503 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commanding Officer
Naval Air Engineering Center
Engineering Specifications and Standards Dept.
(ESSD), Code 93
Lakehurst, NJ 08733

Source: https://assist.dla.mil -- Downloaded: 202640-23T11:16Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-85648 (AS)	AMPLIFIER, TRAVELING WAVE TUBE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:16Z
Check the source to verify that this is the current version before use.